

Technology Stack

Date	01 July 2026
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>HTML5, CSS3, Bootstrap, JavaScript</i>	<i>Provides a responsive, user-friendly interface for entering applicant details and displaying prediction results.</i>

2	Backend / Server-Side	<i>Python, Flask</i>	<i>Flask provides a lightweight web framework to process user requests, integrate the ML model, and return prediction results efficiently.</i>
3	Database / Data Storage	<i>CSV Dataset, Pickle (.pkl) Model</i>	<i>CSV stores historical applicant data for training, while the Pickle file stores the trained Machine Learning model for fast predictions.</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>IBM Watson Machine Learning, IBM Cloud</i>	<i>Enables cloud deployment of the trained model for scalable, secure, and real-time credit card approval predictions.</i>
5	Version Control & CI/CD	<i>Git, GitHub</i>	<i>Supports version control, team collaboration, source code management, and project tracking.</i>

6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib</i>	<i>Used for data preprocessing, feature engineering, model training, evaluation, visualization, and prediction</i>
----------	---	---	--

Additional Technologies Used:

Category	Technology
IDE	VS Code, Jupyter Notebook
Machine Learning	Logistic Regression, Decision Tree, Random Forest, XGBoost
Data Visualization	Matplotlib, Seaborn
Model Serialization	Pickle
Deployment	Flask + IBM Watson Machine Learning